Monolithic Antenna-Coupled Schottky Detector for Millimeter-Wave Power Sensing

Corey Decker, *MIT*, Diego de Lope, *MIT*, and Elijah Haga, *MIT*

***Abstract*—This paper presents the design, fabrication, and characterization of an antenna-coupled Schottky detector structure designed for millimeter-wave RF power sensing in the 60–100 GHz regime. The device integrates a planar dipole antenna with a zero-bias Schottky diode on a p-type Silicon-On-Insulator (SOI) platform. To optimize performance, resistively tapered, dual-dipole, and folded dipole antenna geometries were investigated. We fabricated the device using a streamlined three-mask process, incorporating an aluminum-silicon anneal to form low-resistance contacts. Electrical validation through current-voltage (I–V) characterization confirmed rectifying Schottky behavior across multiple devices. RF characterization was limited by degradation of the aluminum bond pads during dicing and cleaning; therefore, predicted voltage responsivities were estimated from extracted ideality factors rather than directly measured under millimeter-wave illumination.**

***Index Terms*—Schottky diodes, RF power sensors, antenna-diode integration, silicon-on-insulator (SOI), millimeter-wave devices**

I. Introduction

RF power sensors in the millimeter-wave and terahertz bands convert incident electromagnetic power to a measurable DC output without an external bias voltage. Zero-bias Schottky diodes are well suited for this application due to their nonlinear current-voltage characteristic, which rectifies small AC signals at the antenna feed point [1]. Demand for compact, low-cost millimeter-wave sensors is growing across medical imaging, where dielectric contrast between tissue types enables non-ionizing, high-contrast imaging [2]; defense, where short wavelengths enable high-resolution detection and tracking [3]; and emerging 5G-to-6G communications infrastructure, where beamforming systems require precise on-chip power monitoring [4].

Current millimeter-wave Schottky detectors are predominantly realized on III-V substrates such as GaAs, which offer high electron mobility but require specialized foundry processes incompatible with standard silicon fabrication [5]–[7]. RF coupling is typically achieved through waveguide housings or silicon substrate lenses [8]–[10], both of which preclude full chip-scale integration and introduce assembly complexity. Furthermore, in systems where the antenna and diode are fabricated separately, interconnect parasitics at the feed point degrade power transfer efficiency at millimeter-wave frequencies. Together, these constraints limit the accessibility and scalability of existing designs. No prior work has demonstrated monolithic antenna-diode co-integration on a silicon-on-insulator (SOI) substrate using a process compatible with university cleanroom infrastructure.

We report the design, fabrication, and characterization of a micro-meter precision zero-bias Schottky diode, demonstrating that a functional Schottky diode can be realized on a p-type SOI substrate using only a three-mask university cleanroom process. Aluminum was used for the ohmic contacts, while a nickel-gold stack formed the Schottky junction. The fabricated structures were designed to receive millimeter-wave RF power and passively rectify it into a measurable DC voltage. Three antenna topologies were incorporated alongside parametric variation of diode dimensions [Fig. 1], with performance assessed primarily through TLM extraction and DC current-voltage validation.

II. Theory

The RF power sensor rectifies incident millimeter-wave signals using the nonlinear current–voltage characteristic of a zero-bias Schottky diode,

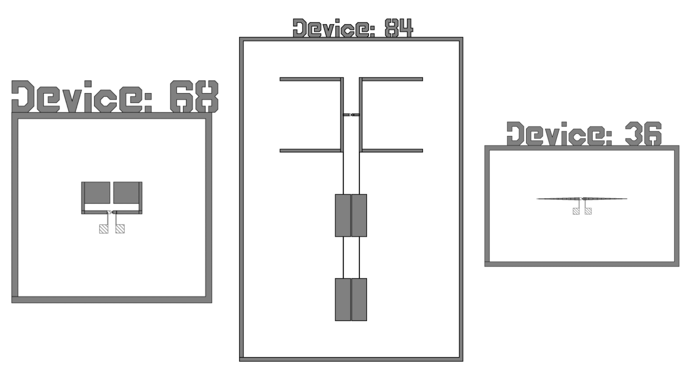

Fig. 1. GDS layout of devices 68, 84, and 36, with folded dipole, dual-dipole, and resistively-tapered antenna topologies, respectively. The antenna arms feed into the central diode region. The two bond pads along the vertical feed line provide probe access to the DC output.

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$$I(V) = I_s \left(e^{V/nV_T} - 1 \right), \tag{1}$$

where I_s is the saturation current, n the ideality factor, and $V_T = k_B T/q$ the thermal voltage. For small RF signals, a Taylor expansion about zero bias yields a quadratic term that produces a DC output proportional to V_{RF}^2. The detector operates in the square-law regime, giving a DC output voltage proportional to incident RF power. The voltage responsivity is

$$\beta = \frac{V_{DC}}{P_{inc}}. \tag{2}$$

At 60–100 GHz, junction capacitance C_j and series resistance R_s limit detector bandwidth through the intrinsic cutoff frequency,

$$f_c = \frac{1}{2\pi R_s C_j}. \tag{3}$$

To ensure efficient detection, device dimensions were varied to increase f_c under fabrication constraints.

III. Experimental

A. Device Structure and Design

The RF power sensor was designed for operation in the 60–100 GHz frequency range, utilizing monolithic integration of a receiving antenna and a rectifying Schottky diode on a p-type SOI substrate featuring a 600 μm silicon handle, a 1 μm buried oxide layer, and a 2 μm moderately doped p-type silicon device layer.

We evaluated three antenna topologies: a resistively tapered dipole for broadband response [6], a dual-dipole configuration for enhanced gain [11], [12], and a folded dipole for compactness and impedance matching [7]. All designs were optimized for direct feed into a zero-bias Schottky diode at the antenna center [13]. To minimize parasitic capacitance and ensure efficient high-frequency operation, the diode was confined to a small active silicon island isolated by the buried oxide layer.

The electrical interface consisted of aluminum bond pads, providing a landing area for needle-probe contacts. A transmission line measurement (TLM) structure at the wafer periphery allowed independent extraction of sheet resistance and contact resistance of the device layer.

B. Fabrication Process

The process proceeded in three photolithographic mask steps, each patterned by direct-write UV lithography [Fig. 2]. The first mask defined and isolated the active silicon device region. Exposed p-type silicon surrounding the island was removed by inductively coupled plasma reactive ion etching (ICP-RIE) with etch conditions selected to produce sloped sidewalls, reducing the risk of metal film discontinuity at the island edge during Schottky contact deposition. Resist was then stripped by plasma ashing. Wet etching approaches, including KOH and 25% TMAH with a silicon nitride hard mask, were evaluated but abandoned due to inadequate etch uniformity.

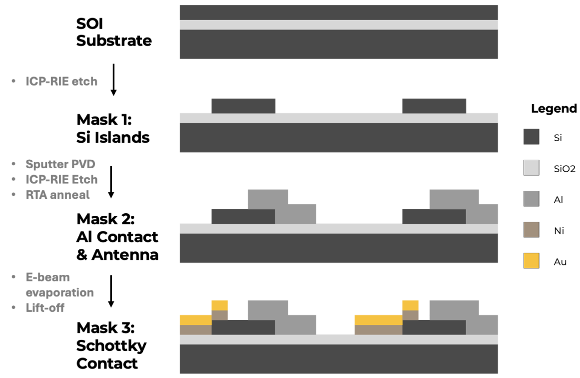

Fig. 2. Cross-sectional three mask fabrication sequence of the RF power sensor on a p-type SOI substrate, with certain process steps outlined. Sloped sidewalls present, but not shown.

The second mask defined the aluminum antenna, traces, bond pads, and ohmic contact region. A 600 nm layer of aluminum was deposited by sputter physical vapor deposition (PVD), patterned by ICP-RIE, and the resist was removed by plasma ashing. The wafer was then annealed at 450°C in a rapid thermal annealer (RTA) for 10 minutes in forming gas, driving an aluminum-silicon reaction beneath the ohmic contact pad to form a heavily doped p$^+$ region and establish a low-resistance ohmic terminal [14].

The third mask defined the Schottky contact by lift-off. A window was opened over the exposed p-type silicon island using a negative-tone image-reversal resist, yielding the undercut profile required for clean lift-off. 150 nm of nickel followed by 250 nm of gold were sequentially deposited by electron-beam evaporation [Fig. 3]. The nickel layer formed the rectifying metal-semiconductor junction at the p-type silicon surface, whereas the gold layer provided oxidation resistance and a high-conductivity contact. Lift-off was performed in acetone, and a light plasma ash removed residual organics without stripping the nickel-gold stack.

Prior to dicing, a protective resist layer was applied to the wafer surface. The wafer was diced into individual chips, after which the protective resist was stripped and a plasma descum removed residual organics.

C. Experimental Characterization

We characterized devices in two stages. Current-voltage (I-V) measurements were first performed at a probe station

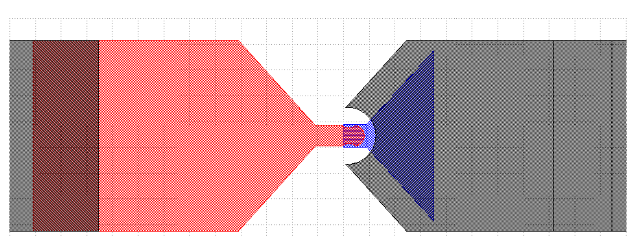

Fig. 3. Close-up of the Schottky contact region showing the Ni/Au anode (red) over the p-type silicon island (blue) and the adjacent aluminum ohmic contact (grey).

using needle contacts on the aluminum bond pads to confirm rectifying behavior of the fabricated Schottky diodes. Second, RF characterization was attempted using a signal source in the 60–100 GHz band to illuminate the device while measuring the rectified DC output voltage across the bond pads. The intended measurement varied source-to-device distance to evaluate the dependence of DC response on incident RF power. However, the RF response could not be quantitatively extracted in this work, as the 150μm GSG probe connected to the power source/measurement unit failed to achieve stable electrical contact due to degraded bond pads.

IV. Results

A. TLM

The TLM structures included in the wafer allowed us to determine the sheet resistance, specific contact resistance, and transfer length. These parameters are essential in determining the efficacy of the ohmic contact on the Schottky diode and its series resistance, which directly limits the cutoff frequency. These variables are related by the following formula:

$$R_T = \frac{R_{Sh} L}{W} + 2R_C \tag{4}$$

where R_T is the total resistance between two pads, R_{Sh} is the sheet resistance, R_C is the contact resistance, and L and W are the dimensions of the TLM. The slope of the R_T–L fitted line, multiplied by the pad width W, yields the sheet resistance R_{Sh}, and the y-intercept yields twice the contact resistance $2R_C$. The transfer length is then

$$L_T = \sqrt{\frac{\rho_C}{R_{Sh}}}, \tag{5}$$

where ρ_C is the specific contact resistance [15]. Extracted values at three annealing conditions are reported in Table I.

TABLE I
TLM Data Under Varying Anneals

Anneal	Sheet Resistance (kΩ/sq)	Specific Contact Resistance (Ω·cm^2)	Transfer Length (μm)
Pre-Anneal	34.63	3.42	99
10 Minute 450°C	7.04	0.55	88
20 Minute 450°C	5.01	1.22	156

The primary effect of annealing is a reduction in specific contact resistance, from 3.42 $\Omega \cdot \text{cm}^2$ pre-anneal to 0.55 $\Omega \cdot \text{cm}^2$ after 10 minutes at 450 °C. The observed variation in sheet resistance across annealing conditions may be attributed to wafer-to-wafer process nonuniformity rather than a true annealing effect, as pre-anneal and post-anneal measurements were performed on different wafers.

B. IV-Characteristics

The electrical performance of the monolithic Schottky diodes was validated through DC current-voltage (I–V) measurements to confirm the rectifying behavior essential for zero-bias millimeter-wave detection. Fig. 4 and Fig. 5 present semilog and linear I–V characteristics for a representative subset of devices spanning the three antenna topologies and the parametric sweep across three design variables: Schottky contact radius r_{sch}, silicon body width w_{body}, and metal-to-silicon gap offset g_{offset}.

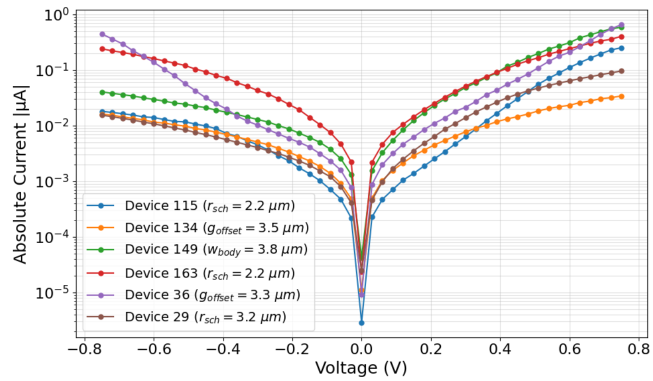

Fig. 4. Semilog current–voltage characteristics of six fabricated Schottky diode devices. All devices exhibit rectifying behavior, with forward current rising exponentially.

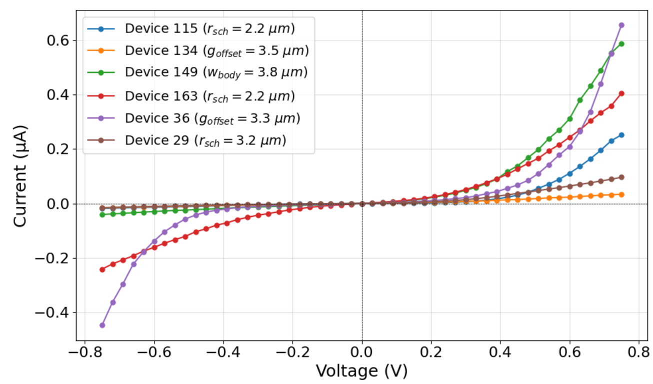

Fig. 5. Linear current–voltage characteristics of six fabricated Schottky diode devices. Current conduction is significantly higher under forward bias compared to reverse bias. This relationship is characteristic of functional Schottky junctions and is essential for signal rectification.

As shown in the semilog plot [Fig. 4], all characterized devices demonstrated nonlinear current-voltage behavior near zero bias, confirming the formation of rectifying Schottky junctions. The current magnitude under both reverse and forward bias varied substantially across the measured devices, suggesting sensitivity to geometry, contact formation, and local process variation.

To quantify the diode behavior more directly, ideality factors n and saturation currents I_s were extracted by fitting the linear region of $\ln(I)$ versus V over 0.1–0.4 V at $T = 293$ K, giving $V_T = 25.26$ mV. Assuming a load resistance of 50 Ω, we predicted voltage responsivities as:

$$\beta_{predicted} = \frac{R_L}{2nV_T}. \tag{6}$$

Extracted parameters and predicted responsivities are summarized in Table II.

The extracted ideality factors range from $n = 3.85$ to 6.42,

TABLE II
EXTRACTED DIODE PARAMETERS AND PREDICTED RESPONSIVITY

Device	n	I_s **(A)**	$\beta_{predicted}$ **(V/W)**
115	3.85	2.95×10^{-10}	257
134	6.42	1.16×10^{-9}	154
149	4.62	3.50×10^{-9}	214
163	5.00	4.53×10^{-9}	198
36	4.84	1.87×10^{-9}	205
29	4.56	9.84×10^{-10}	217

indicating that although the devices behave as rectifying Schottky diodes, their transport is nonideal. Device 115 ($r_{sch} = 2.2\ \mu$m) exhibited the most favorable extracted parameters among the measured subset as it had the lowest ideality factor, the lowest saturation current, and the highest predicted responsivity. By contrast, Device 134 ($g_{offset} = 3.5\ \mu$m) had the highest ideality factor, and therefore the lowest predicted responsivity, consistent with its more restricted forward conduction in the measured I–V curve. Device 149 and Device 36 showed relatively high forward currents, but their larger saturation currents and, in the case of Device 36, elevated reverse leakage affected detector quality.

Because wafer-level process nonuniformity introduced variation in contact formation independent of geometry, the relative contributions of each design variable to the observed spread in n, I_s, and $\beta_{predicted}$ cannot be isolated from the present dataset. Nevertheless, the results indicate that the preferred device must balance strong nonlinearity near zero bias, low leakage, and low effective series resistance.

C. RF Power Sensor

RF characterization was precluded by degradation of the aluminum bond pads during the dicing and cleaning sequence, which reduced pad area below the minimum required for reliable needle-probe contact. Several devices on the diced wafer remain untested, and alternative probing configurations may enable RF characterization without requiring re-fabrication. Future work should additionally increase bond pad dimensions and evaluate alternative protective resist formulations to preserve pad integrity through dicing. Predicted responsivities of 154–257 V/W (Table II), computed assuming a 50 Ω load, provide the expected performance baseline against which future RF measurements should be compared.

V. CONCLUSION

We demonstrated the design, fabrication, and electrical characterization of a monolithic antenna-coupled Schottky detector on a p-type SOI substrate using a three-mask university cleanroom process. The process integrated planar antenna geometries with a Ni/p-Si Schottky junction and annealed aluminum ohmic contacts. TLM characterization showed that a 10-minute aluminum anneal at 450 °C reduced specific contact resistance from 3.42 to 0.55 Ω·cm^2. DC current-voltage measurements of six representative devices confirmed rectifying behavior. Ideality factors extracted from DC I–V measurements ranged from $n = 3.85$ to 6.42, yielding predicted voltage responsivities of 154–257 V/W. RF characterization was prevented by bond pad degradation during dicing; future work should increase pad dimensions, evaluate alternative protective resist formulations, and pursue further RF testing on the existing wafer.

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